

WZ-Plan: A Roadway Work-Zone Testbed for Autonomous Vehicle Motion Planning via Reinforcement Learning

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Abstract—Autonomous driving has made substantial progress under general driving conditions, but roadway work zones remain challenging for motion planning. Work zones temporarily change lane layouts and restrict available road space, complicating interactions with surrounding vehicles and pedestrians, while limited datasets dedicated to work-zone motion planning hinder planner training and evaluation. Simulation supports controlled training and evaluation in safety-critical scenarios, while reinforcement learning allows planners to learn from interactions with the environment. We introduce WZ-Plan, a configurable CARLA–SUMO testbed comprising a normal-traffic work zone and five safety-critical scenario families—curved work zones, constrained corridors, jaywalker encounters, dense traffic, and opposing-lane closures—and demonstrate its use with a jointly trained Proximal Policy Optimization (PPO) planner and structured scene encoder. Scenario-wise evaluation characterizes planner capabilities, failure modes, and within-town transfer limitations, illustrating WZ-Plan’s value as a platform for systematic development and evaluation of work-zone motion planners. <https://shuo9898-zs.github.io/wz-plan/>

I. INTRODUCTION

Autonomous driving has advanced substantially in structured traffic environments, yet reliable operation in safety-critical and long-tail conditions remains challenging. Roadway work zones are particularly demanding because temporary lane configurations, restricted drivable space, traffic-control devices, and interactions with vehicles, workers, and pedestrians may differ substantially from the permanent road structure. In the United States, 905 people were killed in work-zone-related crashes in 2023 and 850 in 2024 [1]. Urban work zones add constrained geometry, intersections, opposing traffic, and limited merging space, producing highly site-specific planning problems.

A study of user-documented automated-vehicle encounters found both successful behavior and persistent challenges involving temporary infrastructure, irregular layouts, and human interaction [2]. However, commercial systems are proprietary, and their public reports do not provide reproducible, scenario-specific success and failure rates. Transparent closed-loop evaluation is therefore needed to determine not only whether a vehicle completes a work zone, but also how and why it fails.

Recent datasets have improved the representation of work zones in autonomous-driving research. ROADWork provides work-zone imagery, scene descriptions, and estimated traversable paths [3]; EPOSH provides ego-view and bird’s-eye-view semantic annotations [4]; and the Waymo End-to-

End Driving Dataset includes construction scenes with observations, routes, and future trajectories [5]. These datasets support perception and offline prediction, but do not provide a controllable environment in which temporary geometry, traffic, and road-user behavior can be varied during repeated closed-loop evaluation.

Reinforcement learning (RL) complements these resources by allowing policies to improve through simulated interaction rather than expert control labels. Applying RL to work-zone motion planning nevertheless requires configurable scenarios, structured observations, and evaluation criteria aligned with safe completion. Proximal Policy Optimization (PPO) offers a practical continuous-control method by constraining policy updates with a clipped objective [6].

We therefore develop a CARLA–SUMO co-simulation testbed for roadway work-zone motion planning. It contains six scenario families spanning conventional and curved work zones, a constrained corridor, jaywalking, dense traffic, and an opposing-lane closure. A structured spatial–temporal encoder represents the ego vehicle, traffic agents, lanes, and temporary work-zone elements, while PPO produces target speed and yaw-rate commands for low-level PID control. We isolate motion planning by assuming structured upstream perception and separate low-level execution, preventing perception and actuation errors from obscuring policy behavior.

The main contributions are:

- WZ-Plan, a configurable CARLA–SUMO testbed for autonomous vehicle motion planning in roadway work zones, comprising a normal-traffic reference scenario and five safety-critical scenario families with varied work-zone geometry and traffic conditions;
- An empirical study of RL-based motion planning in WZ-Plan, using PPO and a structured scene encoder to train a single policy across six scenario families and evaluate its performance and failure modes on new work-zone configurations.

II. RELATED WORK

We review work-zone datasets and evaluation environments, work-zone motion planning, and reinforcement-learning-based driving policies.

A. Work-Zone Datasets and Evaluation Environments

Prior work has largely focused on detecting temporary roadway elements. Shi et al. studied multimodal work-zone detection [7], while Zuo et al. combined multi-source data with topology-based inference for work-zone detection and sizing [8]. EPOSH, ROADWork, and the Waymo

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End-to-End Driving Dataset provide semantic, scene-level, and trajectory annotations [4], [3], [5], and WorkZone3D adds synchronized camera–LiDAR annotations of temporary traffic-control devices [9]. These prerecorded datasets primarily support perception and offline prediction rather than interactive evaluation.

Simulation enables repeatable testing without real-world risk. CARLA provides high-fidelity environments and vehicle dynamics [10], while SUMO provides configurable microscopic traffic [11]. VR-WISE integrates both to study human-driver behavior in work zones [12]. However, closed-loop environments for systematically training and evaluating a common planner across diverse work-zone geometries, traffic conditions, and failure outcomes remain limited.

Work-zone research also addresses trajectory generation. ROADWork provides image-space baselines for navigation-goal and traversable-pathway prediction [3]. Building on it, REACT-Drive combines failure-derived constraints with retrieval-augmented trajectory generation [13], while Work-Drive connects perception-grounded causal reasoning with trajectory prediction [14]. These methods are primarily evaluated offline on prerecorded observations. Game-theoretic work-zone maneuver generation has also been studied [15], but focuses on isolated maneuvers rather than one policy operating across heterogeneous geometries and interactions. These studies leave limited support for evaluating one closed-loop policy across heterogeneous work-zone geometries and interactions.

B. Reinforcement Learning for Autonomous-Vehicle Motion Planning

Driving policies can be learned through imitation or reinforcement learning. Codevilla et al. demonstrated conditional imitation learning in CARLA and on a scaled robotic vehicle [16]. While imitation learning depends on expert-demonstration coverage, RL learns through environment interaction and can directly optimize task outcomes.

Prior driving RL studies improve learning through curriculum design and simplified reward formulations [17], [18]. Other approaches incorporate predicted human attention [19] or neurocognitive reward models [20] to guide safety-critical behavior. These studies demonstrate the flexibility of RL objectives, but generally do not examine one policy across diverse temporary road configurations.

Simulation throughput is also important because RL requires extensive interaction. MetaDrive provides compositional scenarios for generalizable driving RL [21], while GPU-Drive supports massively parallel multi-agent simulation [22]. In contrast, our CARLA–SUMO integration prioritizes detailed work-zone geometry and configurable microscopic traffic, at the cost of lower sampling throughput.

PPO limits large policy changes through its clipped objective [6], making it suitable for continuous motion-planning actions. Nevertheless, RL performance may vary substantially across random seeds and evaluation choices [23]. We therefore evaluate multiple training seeds and report

TABLE I
SCENARIO FAMILIES USED IN THE PROPOSED TESTBED.

ID	Scenario	Map	Primary challenge
S1	Normal traffic	Town05	Standard avoidance
S2	Curved work zone	Town05	Road curvature
S3	Safe corridor	Town10HD	Boundary compliance
S4	Jaywalker	Town10HD	Pedestrian conflict
S5	High traffic	Town02	Limited merging gaps
S6	Oncoming closure	Town02	Opposing traffic

scenario-wise success and mutually exclusive failure outcomes, rather than relying on one aggregate driving score.

III. METHODOLOGY

A. CARLA–SUMO Work-Zone Testbed

1) *Simulator Framework Design*: We develop a CARLA–SUMO–RL framework that separates ego-vehicle control from background-traffic simulation. CARLA [10] executes the PPO-controlled ego vehicle and manages work-zone geometry, collision detection, and scripted jaywalkers, while SUMO [11] controls surrounding traffic. Work zones are represented by forbidden polygons or safe-corridor boundaries with cones and warning signs. Scenarios S1–S3 and S5–S6 use CARLA–SUMO co-simulation, whereas the jaywalker scenario S4 runs entirely in CARLA.

The simulators are synchronized at 10 Hz using a 0.1-s step, with each episode limited to 600 steps. At every step, the ego transform is sent to SUMO as a proxy, allowing SUMO vehicles governed by the Krauss car-following and SL2015 lane-changing models to react to the learned policy. Their updated states and short motion histories are then returned to CARLA and incorporated into the ego observation. This bidirectional synchronization enables closed-loop interactions and allows unsafe ego maneuvers to produce realistic traffic conflicts.

2) *Scenario Selection and Design*: We aim to train a single motion-planning policy across urban work zones varying in temporary geometry, traffic, and vulnerable-road-user interactions. Scenario design draws on MUTCD guidance on channelization and one-lane, two-way operation [24]; studies of curvature, signing, traffic-dependent merging, and compliance [25], [26], [27], [28]; and reported work-zone pedestrian and worker fatalities [1]. Table I summarizes the six scenario families.

Each work-zone placement has three geometric variants. **S1** is the control condition: a conventional obstruction on a straight, bidirectional road with moderate traffic. Its three placements are approximately 8, 16, and 12 m long; the first two have a warning device 8 m upstream, while the third is unsigned. Two SUMO routes use eight initial vehicles, up to 24, and Poisson arrivals at 1.2 vehicles/s. **S2** tests reduced sight distance and mandatory merging on curved roads. Four Town05 placements span clockwise and counterclockwise segments. Two SUMO routes each contain five vehicles without continuous injection. **S3** directs the ego vehicle

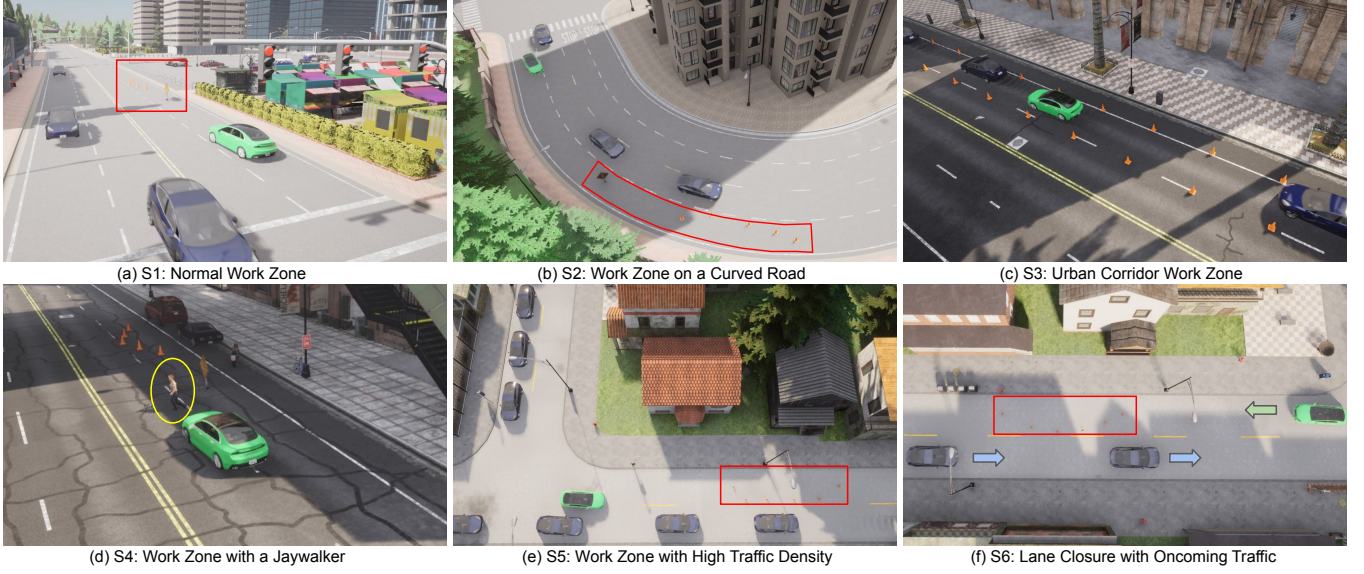


Fig. 1. Overview of the six roadway work-zone scenarios. The green Lincoln denotes the ego vehicle, red boxes highlight the work-zone areas, the yellow ellipse identifies the jaywalker in S4, and arrows in S6 indicate vehicle travel directions. The scenarios vary in road geometry, traffic density, lane configuration, and dynamic hazards.

through cone-defined corridors rather than around lane closures: two 60-m straight corridors, 3.7 and 5.0 m wide, and an approximately 60-m curved corridor with widths of 3.5–6.5 m. Variants alter cone spacing and entrance or exit geometry. SUMO uses 20–40 vehicles with Poisson arrivals at 2.0 vehicles/s, plus a five-vehicle route near the corridor. **S4** models deliberate jaywalking outside designated crossings, using an 8-m signed and a 12-m unsigned placement with pedestrians appearing upstream or downstream. Three crossing events are triggered at ego distances of 45, 30, and 15 m, with pedestrians moving at 2.0 m/s. S4 runs entirely in CARLA without SUMO traffic. **S5** tests lane changing under limited traffic gaps. Its three Town02 placements use 20 initial and up to 40 SUMO vehicles, with Poisson arrivals at 2.0 vehicles/s over four to six routes. **S6** requires temporary entry into the opposing lane and a return after passing the obstruction. Its three Town02 placements generate oncoming two-vehicle platoons with 2.0–2.5-s intra-platoon headways and 30–36-s inter-platoon gaps, with at most four oncoming vehicles active simultaneously.

constructed configurations with different work-zone locations, geometries, and ego origins. Minor controlled differences in traffic and scenario-specific parameters are also permitted while preserving each scenario family’s defining characteristics. Training comprises 18 work-zone placements and 54 geometric origin–layout combinations, with SUMO traffic varying across rollouts. For each scenario, validation and test use separate held-out within-town configurations, each with three ego origins and 10 and 50 rollouts per origin, respectively. Evaluation therefore measures within-town transfer to unseen work-zone configurations. Fig. 1 illustrates the six scenario families.

B. Work-Zone-Aware PPO Motion Planner

1) *Structured Observation and Scene Encoder*: As illustrated in Fig. 2(a), the motion planner receives a fixed-length, ego-centered observation containing the ego state and its history, work-zone elements, dynamic-agent tracks, and local lane segments. The raw 1,311-D vector comprises a 35-D ego representation, up to 12 work-zone elements of 13 dimensions, eight dynamic agents of 28 dimensions, and 128 lane segments of seven dimensions. Each work-zone and agent entry contains an eight-bit episode-local identifier for temporal correspondence. These identifiers are removed before feature learning, leaving 5-D work-zone tokens and 20-D dynamic-agent tokens.

Each modality is processed by a separate token MLP, with weights shared across entries of the same modality. These MLPs map the ego, work-zone, dynamic-agent, and lane inputs to 128-D tokens while preserving their modality-specific representations. The ego token then serves as the query in three masked, eight-head cross-attention modules, whose keys and values are the work-zone, agent, and lane tokens, respectively. This produces one ego-relevant summary for each scene modality while excluding padded entries. A second ego-conditioned cross-attention module uses the ego token to fuse the three modality summaries. Finally, the resulting 128-D fused scene token is concatenated with the 128-D ego token to form the 256-D scene representation.

2) *PPO Training and Parallel Rollout*: As shown in Fig. 2(b), the encoded scene representation is passed to a PPO actor–critic policy implemented in Stable-Baselines3 [29], [6]. The actor and critic use separate feature extractors and two 256-unit Tanh layers. The actor outputs two bounded actions, mapped to target speed $v^* \in [0, 13.89]$ m/s and yaw rate $\omega^* \in [-30, 30]$ deg/s. Longitudinal and lateral PID

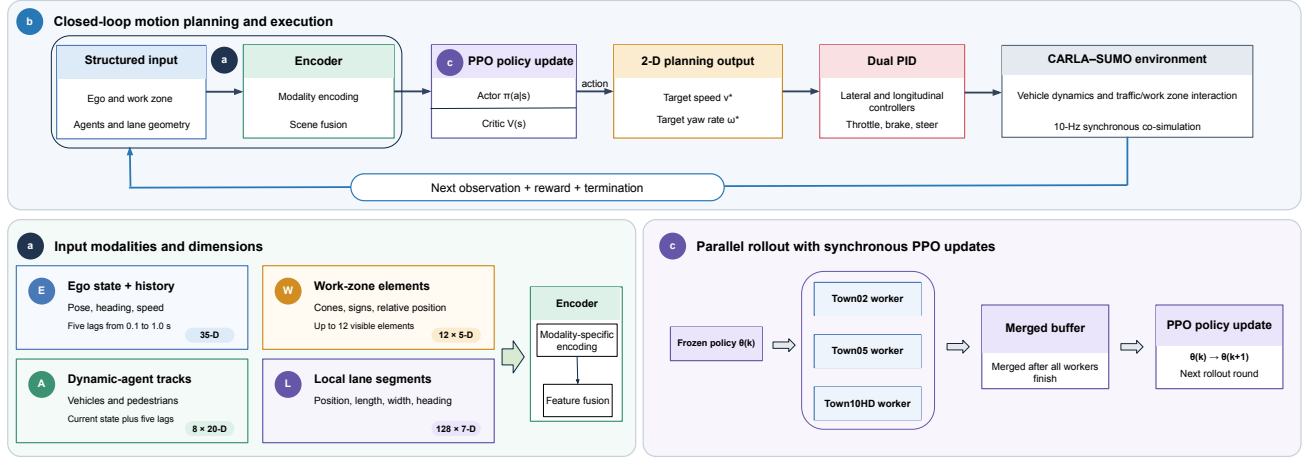


Fig. 2. Overview of the proposed motion-planning and training framework. (a) The structured observation modalities are encoded into a compact scene representation. (b) The PPO policy generates target-speed and target-yaw-rate commands, which are executed by dual PID controllers in the CARLA-SUMO closed loop. (c) Three simulation workers collect parallel rollouts using a frozen policy, and their samples are merged before each synchronized PPO update.

controllers convert these commands into throttle, brake, and steering inputs.

We use a learning rate of 3×10^{-4} , discount factor $\gamma = 0.999$, GAE parameter $\lambda = 0.995$, clipping range $\epsilon = 0.2$, batch size 512, and 20 optimization epochs per update. The value-loss coefficient is 0.5, the entropy coefficient is zero, and the gradient norm is clipped at 0.5. Because PPO is on-policy, each update uses a newly collected rollout buffer. The buffer contains 102,400 valid transitions, comprising 32,400 steps from Town02, 43,000 from Town05, and 27,000 from Town10HD. Training performs 35 policy updates, corresponding to 3,584,000 valid environment steps. Across the 54 training layout-origin combinations, this gives a nominal average of approximately 66,370 transitions per combination, although the realized allocation varies with episode duration and termination.

Rollout collection is the primary computational bottleneck because each transition requires advancing the CARLA-SUMO simulation. We therefore use three independent CPU rollout workers connected to dedicated CARLA servers for Town02, Town05, and Town10HD, as shown in Fig. 2(c). During each rollout round, the workers execute concurrently using the same frozen policy parameters θ_k . After all workers reach their assigned quotas, their transitions are merged into a single on-policy buffer and used to obtain θ_{k+1} . The updated policy is then distributed for the next rollout round. Thus, simulation proceeds in parallel, whereas rollout aggregation and PPO optimization are synchronized.

3) *Reward Design*: The reward combines dense progress with a terminal outcome. Let $\mathbf{o}$ and $\mathbf{g}$ denote the ego origin and finish-line midpoint. For ego position $\mathbf{x}_t$, normalized progress is

$$p_t = \text{clip}\left(\frac{(\mathbf{x}_t - \mathbf{o})^\top (\mathbf{g} - \mathbf{o})}{\|\mathbf{g} - \mathbf{o}\|_2^2}, 0, 1\right). \quad (1)$$

We maintain the high-water mark $\bar{p}_t = \max(\bar{p}_{t-1}, p_t)$, yielding the dense reward $r_t^{\text{prog}} = 50(\bar{p}_t - \bar{p}_{t-1})$. It is computed at every step, prevents reward accumulation through backtracking, and contributes at most 50 per episode. The complete reward is

$$r_t = r_t^{\text{prog}} + d_t r_t^{\text{term}}, \quad d_t \in \{0, 1\}, \quad (2)$$

where d_t indicates termination and exactly one terminal outcome is assigned: success receives +100; collision, work-zone violation, or off-road driving receives -100; and time-out receives -50. A timeout occurs after 600 steps (60 s at 10 Hz). The same reward is used across all six scenarios.

IV. EXPERIMENTS AND RESULTS

A. Training Dynamics

Training was conducted on a workstation with an Intel Core i9-14900KF CPU, 64 GB RAM, and an NVIDIA RTX 4090 GPU, using three parallel three CARLA-SUMO simulator workers. We first examine the training dynamics of joint S1-S6 training across five independent random seeds (7, 17, 27, 37, and 47). Each run comprises 35 PPO training iterations and 3.584×10^6 environment steps. Figure 3 reports the episodic return for each scenario and for all scenarios combined. An episode corresponds to one ego-vehicle trajectory from initialization to termination, including both successful and unsuccessful attempts; its return is the cumulative reward along that trajectory.

The scenario-wise mean returns increase relative to their initial levels, although the extent of improvement differs across scenarios. S1 shows a pronounced increase and maintains positive mean returns over much of the later plotted interval. S3 and S4 also exhibit substantial improvement, with their mean returns reaching values around or above zero. In contrast, S2 improves more gradually and remains at a low return level. S5 and S6 also improve, but their mean returns remain predominantly negative over the displayed

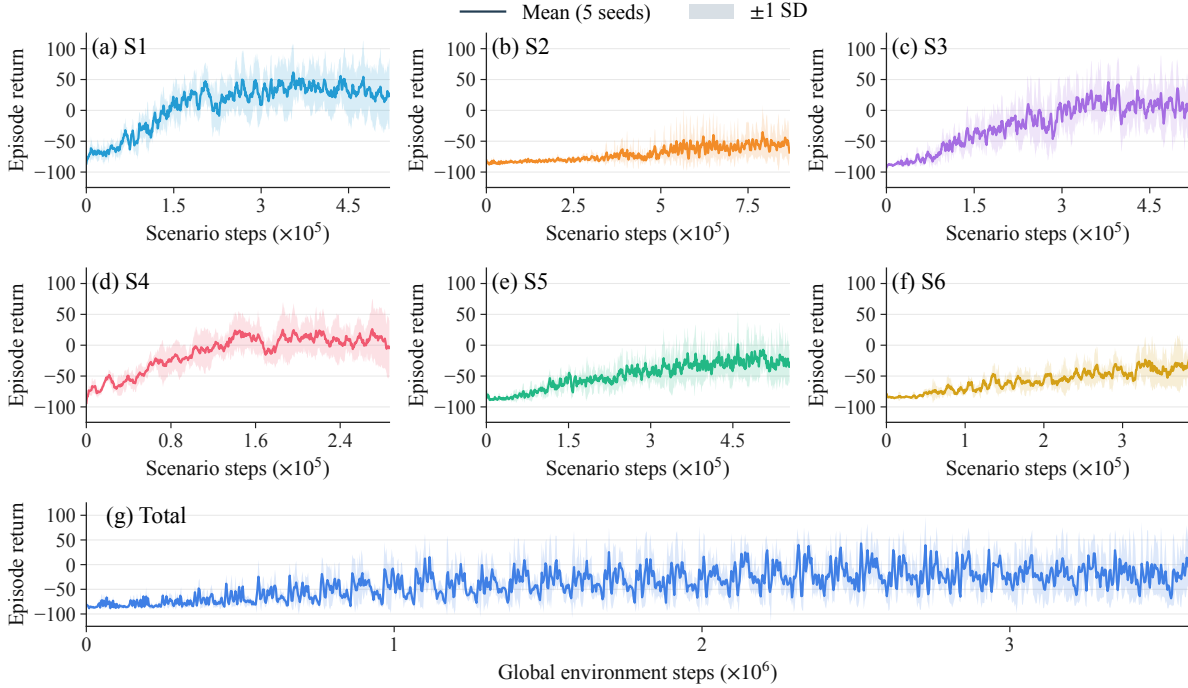


Fig. 3. Joint S1–S6 training across five seeds. Lines and shaded bands show the mean and ± 1 sample standard deviation after per-seed 20-episode smoothing and interpolation onto a common step grid. Panels (a)–(f) use scenario steps; (g) uses global steps. Each panel shows the step range shared by all five seeds.

intervals. The overall return increases markedly during early training, followed by smaller long-term gains and persistent fluctuations.

The shaded bands reveal appreciable variation between training seeds, particularly in S1, S3, and S4. Thus, improved average training return does not imply identical learning trajectories or uniformly stable performance across seeds. These results demonstrate learning progress under joint training while highlighting scenario-dependent behavior and remaining variability. The following subsections complement these training curves with held-out success rates and failure outcomes to assess scenario-specific performance.

B. Limitations on S2 and S6

S2 and S6 exhibit persistently low training returns and inconsistent evaluation performance. Although individual checkpoints attain high validation success on either scenario, these peaks occur separately and coincide with low joint-policy macro performance. We therefore examine a reward modification for S2 and specialist training for S6. Table II reports checkpoints selected by target-scenario validation success.

For S2, both variants use joint S1–S6 training. Road-Arc replaces straight-line progress and removes the terminal speed bonus. It raises validation success from 50% to 90% but does not improve test success, where off-road events dominate both variants (143/150 and 145/150). Because both reward terms change, this single-seed experiment does not isolate road-arc progress alone.

TABLE II
TARGETED S2/S6 SUCCESS RATES (%).

Scenario	Variant	Seed / Policy	Val	Test
S2	Joint / Fixed-OD	7 / P23	50.00	1.33
S2	Joint / Road-Arc	7 / P33	90.00	0.00
S6	S6-only	7 / P32	100.00	0.00
S6	S6-only	27 / P30	53.33	33.33

S6 requires temporary entry into an opposing lane and therefore differs from the lane-following logic of S1–S5. As this condition may be identified from map topology or visual cues, we train two S6 specialists. Despite validation success of 100% and 53.33%, their test success reaches only 0% and 33.33%, with work-zone violations dominating (147/150 and 97/150). Seed27 also varies sharply across test origins (12%, 88%, and 0%).

Thus, both interventions can improve target-scenario validation, but neither resolves transfer to the held-out test configurations.

C. Scenario-wise Success and Failure Analysis

We analyze two fixed policies obtained from joint S1–S6 training. For each training seed, one checkpoint is first selected by S1345 validation success with safety-aware tie-breaking. Among the five resulting checkpoints, Seed27/P32 and Seed17/P27 achieve the two highest macro test success rates, 61.33% and 60.00%, respectively. We retrospectively examine both because their similar macro performance arises

TABLE III
S1345 OUTCOMES OF TWO JOINT POLICIES (%).

Scenario	Val	Test	Collision	Off-road	WZ
Seed27/P32					
S1	100.00	96.67	3.33	0.00	0.00
S3	40.00	16.67	12.67	0.00	70.67
S4	66.67	38.67	0.00	61.33	0.00
S5	83.33	93.33	0.00	6.67	0.00
Macro	72.50	61.33	4.00	17.00	17.67
Seed17/P27					
S1	96.67	100.00	0.00	0.00	0.00
S3	93.33	56.67	6.00	0.00	37.33
S4	30.00	14.00	0.00	85.33	0.67
S5	16.67	69.33	0.00	0.67	30.00
Macro	59.17	60.00	1.50	21.50	17.00

from different scenario-wise outcomes. The policies are evaluated separately, without ensembling or scenario-dependent policy switching.

Table III reports 30 validation episodes and 150 completed test episodes per scenario, with 50 tests per origin. Training, validation, and testing use independently constructed work-zone configurations and ego origins. Collision, Off-road, and WZ are test failure rates computed over all completed test episodes, where WZ denotes a work-zone violation. Simulator-interrupted attempts are excluded and resampled until the required number of completed tests is obtained. Neither policy produces a timeout.

S1: The two policies achieve 96.67% and 100.00% test success. Seed27/P32 obtains 100%, 100%, and 90% success across the three origins, with all five failures caused by vehicle collisions. Seed17/P27 achieves 100% success at every origin. These results are consistent with S1 serving as the baseline scenario, with fewer safety-critical interactions than the other scenario families.

S3: Seed17/P27 achieves 56.67% test success, compared with 16.67% for Seed27/P32. Work-zone violations are the primary failure outcome, accounting for 37.33% and 70.67% of their tests, respectively. The origin-level success rates are 70%, 90%, and 10% for Seed17/P27, and 8%, 12%, and 30% for Seed27/P32, showing substantial variation across starting positions.

S4: Seed27/P32 and Seed17/P27 achieve 38.67% and 14.00% test success, respectively. Neither policy collides with a pedestrian; off-road terminations account for 61.33% and 85.33% of their tests. Visual inspection shows that the policies generally avoid the jaywalker, but some avoidance trajectories eventually leave the road.

S5: Seed27/P32 and Seed17/P27 achieve 93.33% and 69.33% test success. The three origins include different initial-lane conditions, with some requiring a lane change and others allowing the ego vehicle to proceed in its current lane. Their origin-level success rates are 100%/100%/80% and 98%/72%/38%, respectively. The Seed27/P32 failures are off-road terminations, whereas Seed17/P27 fails primarily through work-zone violations.

TABLE IV
SUCCESS ACROSS FIVE JOINT-TRAINING SEEDS (%).

Scenario	Val	Test
S1	84.00 $\pm$ 16.40	66.13 $\pm$ 39.20
S3	68.00 $\pm$ 22.92	38.67 $\pm$ 28.35
S4	64.67 $\pm$ 24.79	51.60 $\pm$ 42.13
S5	41.33 $\pm$ 31.68	52.53 $\pm$ 28.77
Macro	64.50 $\pm$ 8.39	52.23 $\pm$ 9.17

Although the two policies have similar S1345 macro test success, their scenario-wise outcomes differ. Both perform well on S1; Seed17/P27 performs better on S3, whereas Seed27/P32 performs better on S4 and S5. These results characterize the success and failure outcomes of the selected fixed policies rather than establishing an intrinsic scenario difficulty ranking. The next subsection examines performance variation across all five training seeds.

D. Cross-seed and Scenario-wise Variability

We evaluate five policies from independent joint S1–S6 runs. One checkpoint per seed is selected by S1345 validation success with safety-aware tie-breaking. Table IV reports mean $\pm$ sample standard deviation across these policies, rather than rollout-level uncertainty.

The large test deviations for S1 and S4 show that different seeds produce substantially different policies under the same joint-training configuration. This variation is scenario dependent: Seed37/P26 achieves 2.67% on S1 and 97.33% on S4, whereas Seed7/P28 achieves 61.33% and 12.67%, respectively. These contrasting performance profiles show that the macro average can obscure substantial scenario-specific differences among policies.

Training, validation, and test use independently constructed configurations with different work-zone locations, geometries, and ego origins, together with minor differences in traffic and scenario parameters. Thus, validation–test gaps reflect transfer across configurations and finite-sample rollout variability, rather than repeated evaluation of the same geometry. The largest mean decrease occurs in S3 (68.00% to 38.67%), while S5 increases from 41.33% to 52.53%.

Variation also occurs within a run. From Seed27/P31 to P32, S3 validation success decreases from 100% to 40%, while S4 increases from 3.33% to 66.67%. Cross-run and training-time variability is recognized in RL [23]. Stochastic on-policy training and interactive traffic are plausible contributors, although their individual effects are not isolated here.

E. Ablation Studies and Design Choices

We finally summarize the development experiments that motivated the adopted structured-encoder PPO policy. Table V reports the main comparisons. These experiments were conducted during system development and do not all share an identical training budget; they are therefore used to justify design choices rather than to establish universal algorithm rankings.

TABLE V
SUMMARY OF POLICY-DESIGN EXPERIMENTS.

Component	Alternative	Main observation
Encoder	Flat MLP	Almost no successful training episodes
Algorithm	Three SAC variants	Training success of 0.03–0.18%
PPO rollout	Smaller buffer	Lower and less stable seen-scene performance
Training set	S1–S5 only	Lower S1345 validation in all three matched seeds
Reward	Road-arc progress	Higher S2 validation but no test improvement

The Flat-MLP baseline directly maps the 1,311-D observation through separate two-layer actor and critic networks with 256 hidden units per layer. It contains approximately 0.80 million trainable parameters, compared with 1.10 million for our structured-encoder PPO policy. Thus, the two models have comparable parameter counts, and the observed difference is unlikely to be explained solely by model capacity. Nevertheless, the Flat MLP produces almost no successful episodes during training, whereas the structured encoder consistently yields successful policies. This result indicates the importance of explicitly preserving the typed spatial and temporal structure of the driving scene.

The three SAC pilots likewise produced only 0.03%, 0.15%, and 0.18% successful training episodes and were therefore not advanced to test evaluation. A historical PPO configuration using 34,134 transitions per update for 60 updates reached 29.32% in its best seen-scene evaluation and fell to 15.12% at the final checkpoint. We consequently retained the larger 102,400-transition rollout used by the main experiments. Because these PPO configurations also differ in total training budget and update settings, this result supports the adopted training schedule but does not isolate buffer size alone. Finally, matched-seed comparisons show that joint S1–S6 training improves S1345 validation success over S1–S5-only training by 22.50, 15.83, and 15.83 percentage points for Seeds 7, 27, and 47, respectively. This suggests that the additional interaction patterns in S6 provide useful training diversity even though S6 itself remains difficult. The S2 road-arc reward experiment is reported in Section IV-B; its validation improvement did not transfer to the held-out test configuration. Together, these observations motivate the structured encoder, PPO, large-rollout schedule, and joint S1–S6 training used for the reported policy.

V. DISCUSSION, LIMITATIONS, AND CONCLUSION

Discussion. S2 and S6 reveal distinct challenges. S6 requires the ego vehicle to temporarily enter an opposing lane, conflicting with the lane-following behavior shared by the other scenarios. Although S6-specific policies occasionally learn this behavior, their validation–test gaps indicate weak transfer. Since such closures may be identified from map topology or vision, a gated specialist could be more appropriate. S2 is unexpectedly difficult because it preserves normal traffic

logic and primarily introduces curvature. Replacing straight-line progress with road-arc progress improves validation but not test performance, suggesting that reward alignment alone is insufficient. Route-aligned coordinates, explicit curvature features, and more diverse curved-road training may be needed.

The same policy nevertheless handles the nontrivial interactions in S1, S3, S4, and S5. S1 is relatively the most accessible, while S3 demonstrates temporary-corridor following despite occasional boundary violations. In S4, the ego sometimes enters an empty opposing lane to avoid a jaywalker when stopping or crossing the centerline are the only feasible responses. Other attempts leave the roadway, possibly because the timeout penalty discourages prolonged stopping; this hypothesis requires a reward ablation. S5 remains challenging because dense SUMO traffic reduces acceptable gaps while the closure forces a lane change. Joint S1–S6 training also outperforms S1–S5-only training across three matched seeds, possibly because S6 provides additional interaction diversity.

Limitations. Performance varies substantially across seeds and checkpoints within the same run. Improvements in one scenario can also coincide with degradation in another. PPO’s on-policy sampling [6] and stochastic CARLA–SUMO interactions are plausible contributors, but their effects have not been isolated. Training is also costly: each 3.584-million-step joint run requires approximately 15 h despite three parallel simulator workers, with rollout collection forming the main bottleneck. Lightweight or GPU-parallel simulators may improve sampling throughput, subject to different fidelity and traffic-integration trade-offs.

Each scenario currently contains one held-out test work zone with three origins. Fifty rollouts per origin characterize traffic stochasticity but cannot replace additional independent geometries. The results therefore demonstrate within-town transfer across independently constructed training, validation, and test configurations, rather than cross-town or real-world generalization. Furthermore, the evaluation emphasizes safe completion and does not yet quantify comfort, efficiency, or effects on surrounding traffic.

Conclusion. We presented a CARLA–SUMO–RL framework for motion planning across heterogeneous roadway work zones. A structured encoder and joint PPO training enable one policy to handle conventional closures, temporary corridors, pedestrian conflicts, and dense-traffic merging. Curved roads, opposing-lane closures, and training variability remain important generalization boundaries. Future work will improve training stability, expand held-out and cross-town evaluation, and investigate route-aware representations and gated specialists.

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